DESIGN SYSTEMS

TECHNICAL DATA SHEET PREPOX A+B
TRANSPARENT EPOXY FORMULATION

DESCRIPTION

PREPOX A+B is an extremely versatile, solvent-free, two-component epoxy formulation based on epoxy resins and a cycloaliphatic amine hardener. A consolidating primer for concrete with excellent reactivity and excellent impermeability, which can be filled with aggregates for producing epoxy screeds.

It has good penetrating and consolidating power on concrete substrates. The particular chemical structure of the amine hardener ensures good reactivity of the system even at low temperatures.

USES

- Primer for concrete, ideal for promoting adhesion and producing multi-layer coatings, applied by roller or as a skim coat.
- Bonding primer and binder for producing epoxy screeds.
- Impregnation of glass fibres for producing fibreglass coatings.
- Consolidation of reinforced concrete structures and filling of cavities.

SUBSTRATE

The substrate must have a minimum compressive strength of 25 N/mm^2 and a minimum tensile strength of 1,5 N/mm^2.

SUBSTRATE PREPARATION

When working on concrete substrates, check that there is no rising damp. If the concrete is newly built, wait for it to cure completely. The surface must be solid, absorbent and free from oils, surfactants, water and dust; any weak parts must be removed. Floors must be treated mechanically by grinding, shot-blasting or milling.

APPLICATION

At the time of application, combine part A and part B in a single container and mix carefully for 2 minutes using suitable equipment (drill with mixing paddle). Use the entire contents of the container quickly. When emptying the container, avoid scraping the edges and the bottom, as there may be product that is not perfectly blended.

PREPOX can be applied in several ways:

- as a consolidant: apply by roller or brush, neat or diluted with 5-10% ethyl alcohol or universal solvent;
- for simple skim coats and multi-layer coatings: apply as a skim coat, filled with 50% QUARTZ B0 or B1;
- for epoxy screeds: apply as a thick layer, filled with QUARTZ MIX1 or MIX3;
- as a repair mortar: fill the product with thickener up to 10% by weight.

Consumption varies considerably according to the application method and the condition of the substrate: for precise guidance, contact the Design Systems technical service.

WARNINGS

For low-temperature applications, the material can be warmed to 25°C to facilitate application and curing (reduction in viscosity).

PRODUCT DATA

DESCRIPTION	RESULT
Colour	Transparent
Consumption	By roller: 0,100 kg/m^2 As skim coat: 0,350 kg/m^2 of A+B + 0,175 kg/m^2 of QUARTZ B0
Specific gravity (at 25°C)	A+B: 1,10 ± 0,05 g/ml A+B filled: 1,50 ± 0,10 g/ml (with 50% QUARTZ B0)
Viscosity (at 25°C)	A+B: 470 ± 100 mPa·s (spindle 2, 60 rpm) A+B filled: 840 ± 150 mPa·s (with 50% QUARTZ B0, spindle 3, 60 rpm) Diluted 10% with universal solvent: 175 ± 40 mPa·s (spindle 1, 30 rpm)
Solids content (A+B)	100% (approx.)
VOC ready for use (Italian Legislative Decree 161/06)	< 200 g/l – Cat. A/j: two-component high-performance paint (SB)
Flash point	> 100°C
Tool cleaning	Universal solvent
Storage	12 months, in a dry place at a temperature between 5°C and 35°C

APPLICATION DATA AND TIMES

DESCRIPTION	RESULT
Mixing ratio (by weight)	A = 100, B = 50
Pot life (50% R.H.)	at 10°C: > 60 min at 25°C: > 30 min at 30°C: > 20 min
Touch dry (50% R.H.)	at 10°C: 12-16 hours at 25°C: 5-7 hours at 30°C: 2-3 hours
Foot traffic (50% R.H.)	at 25°C: 12 hours
Overcoating (50% R.H.)	at 25°C: 12 to 36 hours
Full traffic (50% R.H.)	at 25°C: 36 hours
Full-depth curing (50% R.H.)	at 25°C: 7 days
Ambient conditions for use	Temperature between +10°C and +30°C, R.H. < 60%, substrate moisture < 4% (*)

TECHNICAL PERFORMANCE DATA

DESCRIPTION	RESULT
Appearance	Glossy

Compressive strength (UNI 4279)	60 N/mm^2
Flexural strength (UNI 7219)	40 N/mm^2
Hardness (ASTM D 2240)	Shore D 78
BCA wear resistance (EN 13892-4)	Class AR 0,5 (10 mm)
Bond strength (EN 13892-8)	4,0 N/mm^2
Chemical resistance	Contact the Design Systems technical service

(*) PREPOX must be applied with a substrate temperature of not less than 15°C and at least 3°C above the dew point.

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof, according to Design Systems systems. The methods applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. However, for uses other than those indicated, preliminary tests are recommended. In any case, Design Systems Srl accepts no responsibility for damage or defects caused by the use of its products, since the conditions of use are not under its control. It also advises that its technical service is available to customers for any information concerning the correct use of its products.